

CS471 – Web Technologies (Laboratory)		Lab 10
		Django Forms (Part 1)

This lab session covers Django Forms (Part 1 – Create, read (display or list), update, and delete). By the end of this session, students should have a very good understanding of how to build CRUD operations in Django.

Pre-lab Preparation:

1. The table book as in lab7.
2. The fake data from mockaroo.com website or any other methods.

Lab Activities:

Part 1: CRUD operations and Template organization (without using Django forms)

Task 1: Create a URL '/books/lab9_part1/listbooks' with any necessary HTML file and view function to list books and provide a link to add a new book (implemented in the following task (Task 2)). Also, two buttons/links are associated with each book that hold the book's ID to edit or delete the book (implemented in Task 3 and Task 4).

Task 2: Create a URL '/books/lab9_part1/addbook' with any necessary HTML file and view function to insert the data of a new book.

Task 3: Create a URL '/books/lab9_part1/editbook/<id>' with any necessary HTML file and view function to update the data of the selected book.

Task 4: Create a URL '/books/lab9_part1/deletebook/<id>' with a view function to delete the book with the specified id.

دالة عرض الكتب 1.

```
def lab9_part1_listbooks(request):
    books = Book.objects.all()
    return render(request, 'bookmodule/lab9_part1/listbooks.html', {'books': books})
```

دالة إضافة كتاب 2.

```
def lab9_part1_addbook(request):
    if request.method == "POST":
        new_title = request.POST.get('title')
        new_price = request.POST.get('price')
        Book.objects.create(title=new_title, price=new_price)
        return redirect('lab9_part1_listbooks')
    return render(request, 'bookmodule/lab9_part1/addbook.html')

# 3. دالة تعديل كتاب
def lab9_part1_editbook(request, id):
    book = Book.objects.get(id=id)
    if request.method == "POST":
        book.title = request.POST.get('title')
        book.price = request.POST.get('price')
        book.save()
        return redirect('lab9_part1_listbooks')
    return render(request, 'bookmodule/lab9_part1/editbook.html', {'book': book})

# 4. دالة حذف كتاب
def lab9_part1_deletebook(request, id):
    book = Book.objects.get(id=id)
    book.delete()
    return redirect('lab9_part1_listbooks')
```

Part 2: CRUD operations with validation (using Django forms)

Task 1: Reimplement all tasks in Part 1 using Django forms with URLs starting as '/books/lab9_part2/...'

```
# دالة الإضافة باستخدام Django Forms
def lab9_part2_addbook(request):
    if request.method == "POST":
```

```
form = BookForm(request.POST)
if form.is_valid():
    form.save()
    return redirect('lab9_part1_listbooks')
else:
    form = BookForm()
    return render(request, 'bookmodule/lab9_part2/addbook.html', {'obj': form})

# دالة التعديل باستخدام Django Forms
def lab9_part2_editbook(request, id):
    book = Book.objects.get(id=id)
    if request.method == "POST":
        form = BookForm(request.POST, instance=book)
        if form.is_valid():
            form.save()
            return redirect('lab9_part1_listbooks')
    else:
        form = BookForm(instance=book)
    return render(request, 'bookmodule/lab9_part2/editbook.html', {'obj': form})
```

My Library - Books List

[+ Add New Book](#)

ID	Title	Price	Actions
1	Tax Accountant	38.0	Edit Delete
2	Senior Editor	99.0	Edit Delete
3	Desktop Support Technician	39.0	Edit Delete
4	Office Assistant IV	97.0	Edit Delete
5	Research Assistant IV	77.0	Edit Delete
6	Payment Adjustment Coordinator	51.0	Edit Delete
7	Analyst Programmer	53.0	Edit Delete
8	Help Desk Technician	96.0	Edit Delete
9	Account Representative III	5.0	Edit Delete
10	Graphic Designer	95.0	Edit Delete
11	Programmer Analyst III	13.0	Edit Delete
12	Senior Developer	90.0	Edit Delete
13	Junior Executive	51.0	Edit Delete
14	Teacher	53.0	Edit Delete
15	Design Engineer	92.0	Edit Delete
16	Food Chemist	14.0	Edit Delete
17	Analyst Programmer	67.0	Edit Delete
18	Nurse	70.0	Edit Delete
19	Legal Assistant	54.0	Edit Delete
20	Editor	98.0	Edit Delete

Image Model - Tea...

Add a New Book

Book Title:

Price:

[Cancel](#)

Edit Book: Tax Accountant

Update Title:

Update Price:

[Cancel](#)